

Ingegneria Informatica e Robotica
Information and Estimation Theory

DETECTION THEORY

Luca Rugini

A.A. 2025/26



Content

1. Detection Theory

   - 1.1 Binary Hypothesis Testing
   - 1.2 Neyman-Pearson Criterion
   - 1.3 Receiver Operating Characteristic (ROC)
   - 1.4 Minimum Probability of Error
   - 1.5 Bayes Risk
   - 1.6 Multiple Hypothesis Testing
   - 1.7 Generalized Likelihood Ratio Test (GLRT)



References

- **Textbook**

  S. Kay, Fundamentals of Statistical Signal Processing Vol. II: Detection Theory, Prentice Hall, 1998

- **These slides**

  Available in the Unistudium repository

- **Additional books**

  B. Levy, Principles of Signal Detection and Parameter Estimation, Springer, 2008

  P. Moulin, V. V. Veeravalli, Statistical Inference for Engineers and Data Scientists, Cambridge University Press, 2018



PART 1

Detection Theory

Motivation

- Given a set of noisy data (observations), **a detector tries to identify if an event of interest occurs or not**, using signal processing techniques

- The applications are many; some of them are
  - Radar
  - Communications
  - Speech
  - Sonar
  - Image processing
  - Biomedicine
  - Control
  - Seismology

What is a detector?

- From an observed data vector $\mathbf{x} = [x[0], \ldots, x[N-1]]^T$, a detector decides between two hypotheses H_0 and H_1 ➔ this is a **binary hypothesis test**

- This decision is done by exploiting the two pdfs of the data vector $\mathbf{x}$ under the two hypotheses

$$f_x(\mathbf{x}; H_0) \qquad f_x(\mathbf{x}; H_1)$$

- A detector often uses a scalar function $T(\mathbf{x})$ of the data and a scalar threshold γ, both computed using the above pdfs



What is a detector?

- Usually the decision is made by comparing the test function $T(\mathbf{x})$ with the threshold γ

- Example:

$$\text{if} \quad T(\mathbf{x}) > \gamma, \quad \text{then decide} \quad \hat{H} = H_1,$$

$$\text{otherwise decide} \quad \hat{H} = H_0$$

- The test T and the threshold γ can be chosen according to some performance criteria



Types of detection problems

- A detection problem can be either a **binary** hypothesis test or a **multiple** hypothesis test
- A detection problem can be either a **parametric** hypothesis test or a **non-parametric** hypothesis test
- A detection problem can be either a **simple** hypothesis test or a **composite** hypothesis test
- A detection problem can be either **deterministic** or **Bayesian**

Example: binary hypothesis testing

- Often the detection problem is a hypothesis test between **two hypotheses** H_0 (null hypothesis) and H_1 (alternative hypothesis)

- Example: the pdf of a sample $x = A + w$ is given by

$$f_x(x;A) = \frac{1}{\sqrt{2\pi\sigma_w^2}} \exp\left[-\frac{(x-A)^2}{2\sigma_w^2}\right]$$

where the parameter A can be either 0 or 1 ➔ this is a **binary** hypothesis testing problem:

$$\boxed{\begin{aligned} H_0&: \ A = 0 \\ H_1&: \ A = 1 \end{aligned}}$$

Example: multiple hypothesis testing

- Sometimes the detection problem is a hypothesis test among **more than two hypotheses**

- Example: the pdf of a sample $x = A + w$ is given by

$$f_x(x;A) = \frac{1}{\sqrt{2\pi\sigma_w^2}} \exp\left[-\frac{(x-A)^2}{2\sigma_w^2}\right]$$

where A can be either -3 or -1 or 1 or 3 ➔ this is a **multiple** hypothesis testing problem:

$$\begin{aligned} H_0&: \ A = -3 \\ H_1&: \ A = -1 \\ H_2&: \ A = 1 \\ H_3&: \ A = 3 \end{aligned}$$

Example: parametric hypothesis testing

- A hypothesis test is **parametric** when **the pdfs** of the data $f_x(x;H_0)$ and $f_x(x;H_1)$ have the same shape and **differs by a parameter only**

- Example: the pdf of a sample $x = A + w$ is given by

$$f_x(x;A) = \frac{1}{\sqrt{2\pi\sigma_w^2}} \exp\left[-\frac{(x-A)^2}{2\sigma_w^2}\right]$$

where the parameter A can be either 0 or 1 ➔ this is a **parametric** hypothesis testing problem:

$$\boxed{\begin{aligned} H_0&: \ A = 0 \\ H_1&: \ A = 1 \end{aligned}}$$

Example: non-parametric hypothesis testing

- A hypothesis test is **non-parametric** when **the pdfs** of the data $f_x(x;H_0)$ and $f_x(x;H_1)$ have **different shapes**

- Example: the pdf of $x[n] = w[n]$ is given by

$$f_x(x;H_0) = \frac{1}{\sqrt{2\pi\sigma_w^2}} \exp\left[-\frac{(x-\mu_w)^2}{2\sigma_w^2}\right] \qquad f_x(x;H_1) \neq \frac{1}{\sqrt{2\pi\sigma_w^2}} \exp\left[-\frac{(x-\mu_w)^2}{2\sigma_w^2}\right]$$

➔ this is a **non-parametric** hypothesis testing problem:

$$\begin{aligned} &H_0\text{: } w \text{ is Gaussian} \\ &H_1\text{: } w \text{ is not Gaussian} \end{aligned}$$

Example: simple hypothesis testing

- A hypothesis test is **simple** when **the pdfs** of the data $f_x(x;H_0)$ and $f_x(x;H_1)$ **are completely known**

- Example: the pdf of a sample $x = A + w$ is given by

$$f_x(x;A) = \frac{1}{\sqrt{2\pi\sigma_w^2}} \exp\left[-\frac{(x-A)^2}{2\sigma_w^2}\right]$$

where the parameter A can be either 0 or 1 ➔ this is a **simple** hypothesis testing problem:

$$\boxed{\begin{aligned} H_0&: \ A = 0 \\ H_1&: \ A = 1 \end{aligned}}$$

Example: simple hypothesis testing

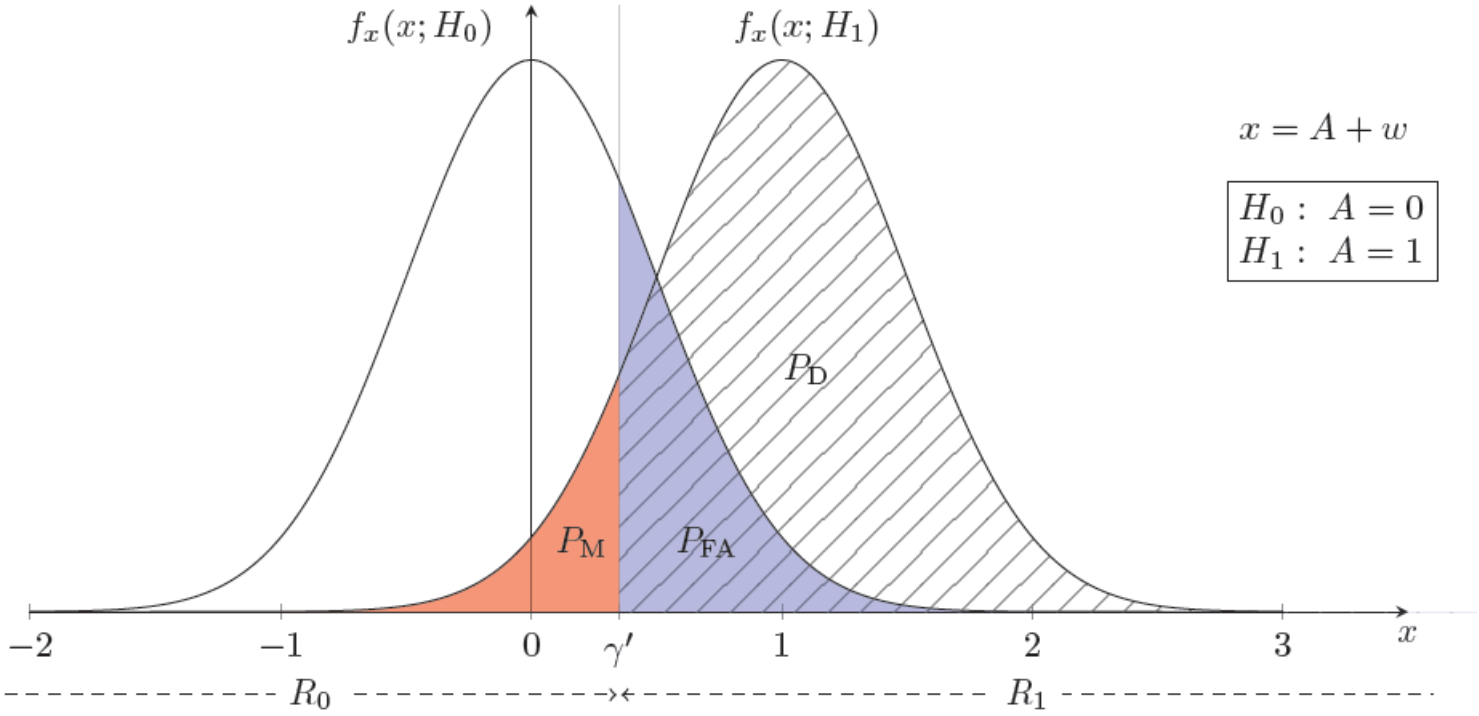


Figura 5.13: Esempio di HT binaria, parametrica, semplice e deterministica.

Example: composite hypothesis testing

- A hypothesis test is **composite** when **at least one of the pdfs** $f_x(x;H_0)$ and $f_x(x;H_1)$ **is partially unknown**

- Example: the pdf of a sample $x = A + w$ is given by

$$f_x(x;A) = \frac{1}{\sqrt{2\pi\sigma_w^2}} \exp\left[-\frac{(x-A)^2}{2\sigma_w^2}\right]$$

where the parameter A can be either 0 or nonzero

➔ this is a **composite** hypothesis test:

$$\boxed{\begin{aligned} H_0&: \ A = 0 \\ H_1&: \ A \neq 0 \end{aligned}}$$

Example: composite hypothesis testing

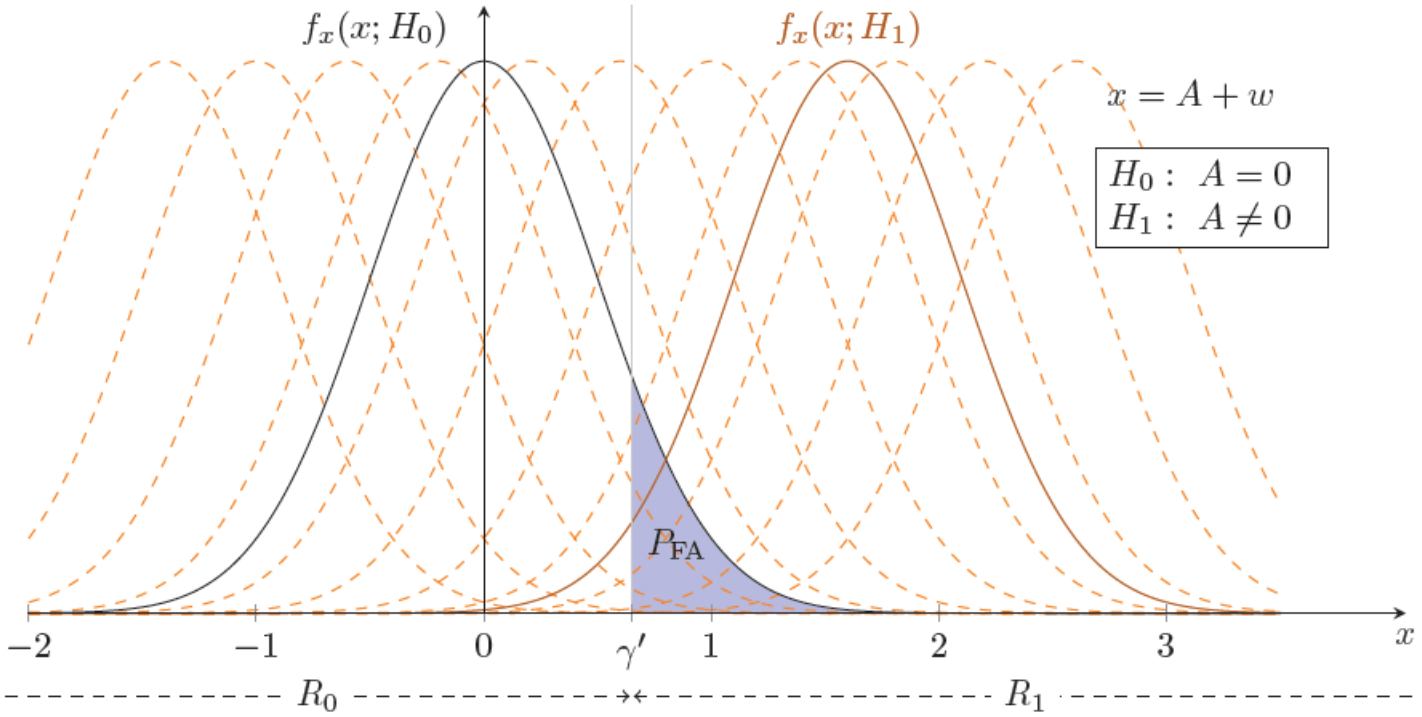


Figura 5.14: Esempio di HT binaria, parametrica, composta e deterministica.

Example: deterministic hypothesis testing

- A detection problem is **deterministic** when there is **no probabilistic knowledge** about the occurrence of the hypotheses: $\Pr\{H_0\}$ and $\Pr\{H_1\}$ are unknown or undefined

- Example: the pdf of a sample $x = A + w$ is given by

$$f_x(x;A) = \frac{1}{\sqrt{2\pi\sigma_w^2}} \exp\left[-\frac{(x-A)^2}{2\sigma_w^2}\right]$$

where A can be either 0 or 1, but $\Pr\{A=0\} = \Pr\{H_0\}$ and $\Pr\{A=1\} = \Pr\{H_1\} = 1 - \Pr\{H_0\}$ are unknown:

$$\boxed{\begin{aligned} H_0&: \ A = 0 \\ H_1&: \ A = 1 \end{aligned}}$$

Example: Bayesian hypothesis testing

- A detection problem is **Bayesian** when there is **some probabilistic knowledge** about the occurrence of the hypotheses: $\Pr\{H_0\}$ and $\Pr\{H_1\}$ are known in advance

- Example: the pdf of a sample $x = A + w$ is given by

$$f_x(x;A) = \frac{1}{\sqrt{2\pi\sigma_w^2}} \exp\left[-\frac{(x-A)^2}{2\sigma_w^2}\right]$$

where A can be either 0 or 1, with $\Pr\{A=0\} = \Pr\{H_0\}$ and $\Pr\{A=1\} = \Pr\{H_1\} = 1 - \Pr\{H_0\}$ known a priori:

$$\boxed{\begin{aligned} H_0&: \ A = 0 \\ H_1&: \ A = 1 \end{aligned}}$$

Example: Bayesian binary hypothesis testing

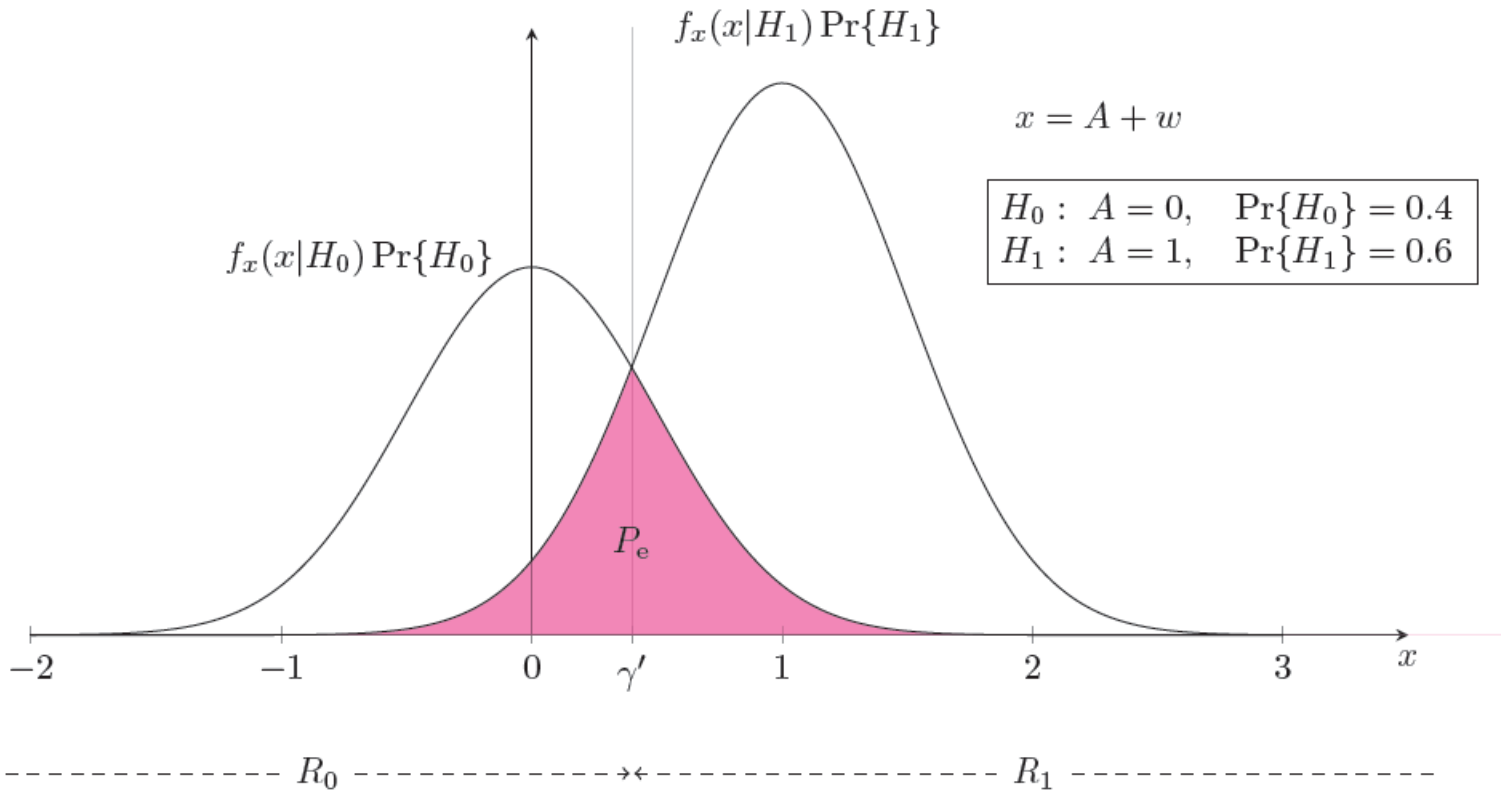


Figura 5.15: Esempio di detector MAP per HT binaria, parametrica, semplice e bayesiana.

Example: Bayesian multiple hypothesis testing

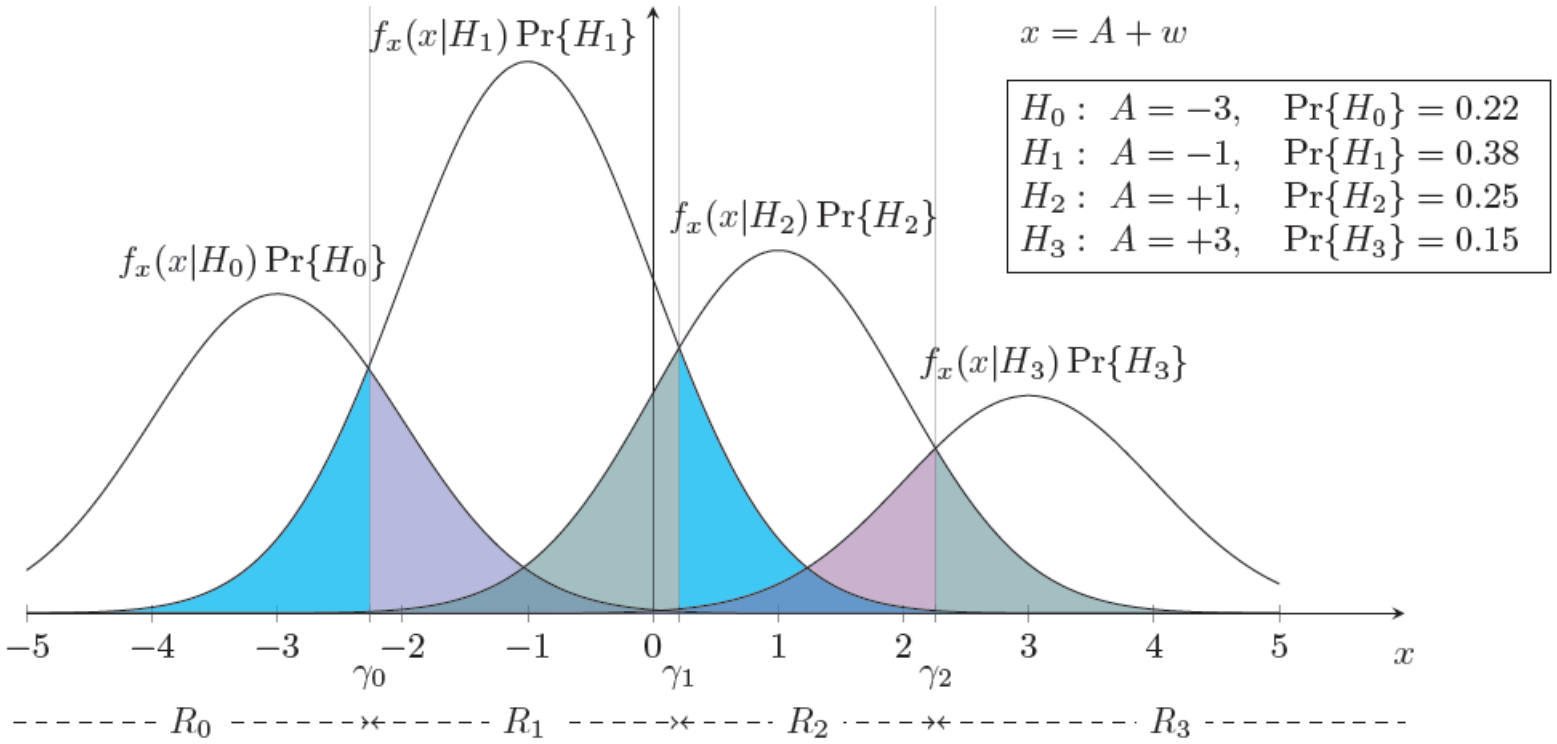


Figura 5.17: Esempio di detector MAP per HT multipla, parametrica, semplice e bayesiana.

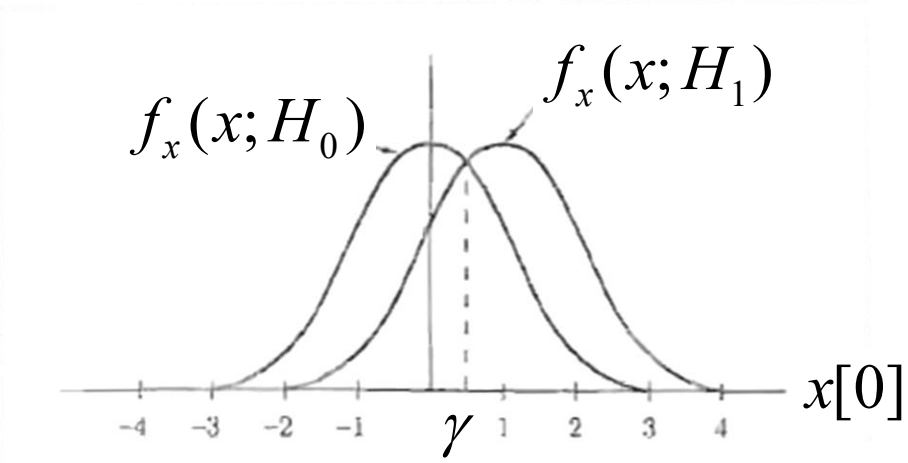


1.1

Binary Hypothesis Testing

Binary hypothesis testing

- Example of simple binary hypothesis testing (single sample)

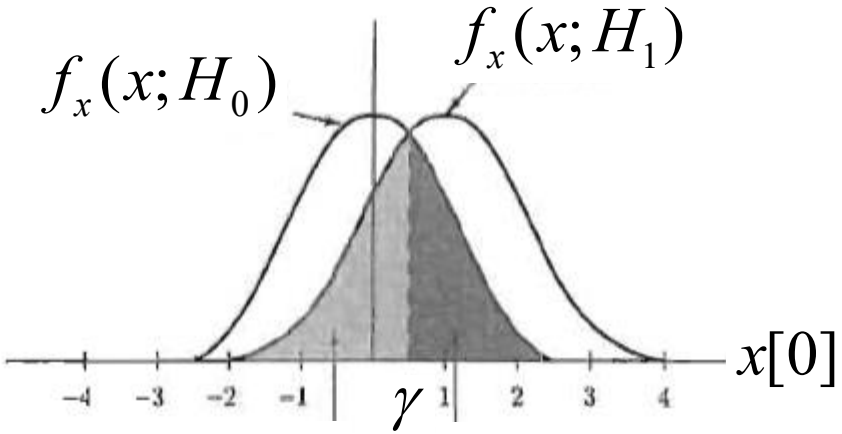


$$H_0: \quad x[0] = w[0]$$
$$H_1: \quad x[0] = 1 + w[0]$$

- The goal is to identify two regions ➔ the detector decides $\hat{H}_0$ when $x[0] \in R_0 = \{x: \quad x[0] < \gamma\}$, while decides $\hat{H}_1$ when $x[0] \in R_1 = \{x: \quad x[0] > \gamma\}$

Binary hypothesis testing

- There are two types of errors: a miss detection (light grey) and a false alarm (dark grey)



$$H_0: \quad x[0] = w[0]$$
$$H_1: \quad x[0] = 1 + w[0]$$

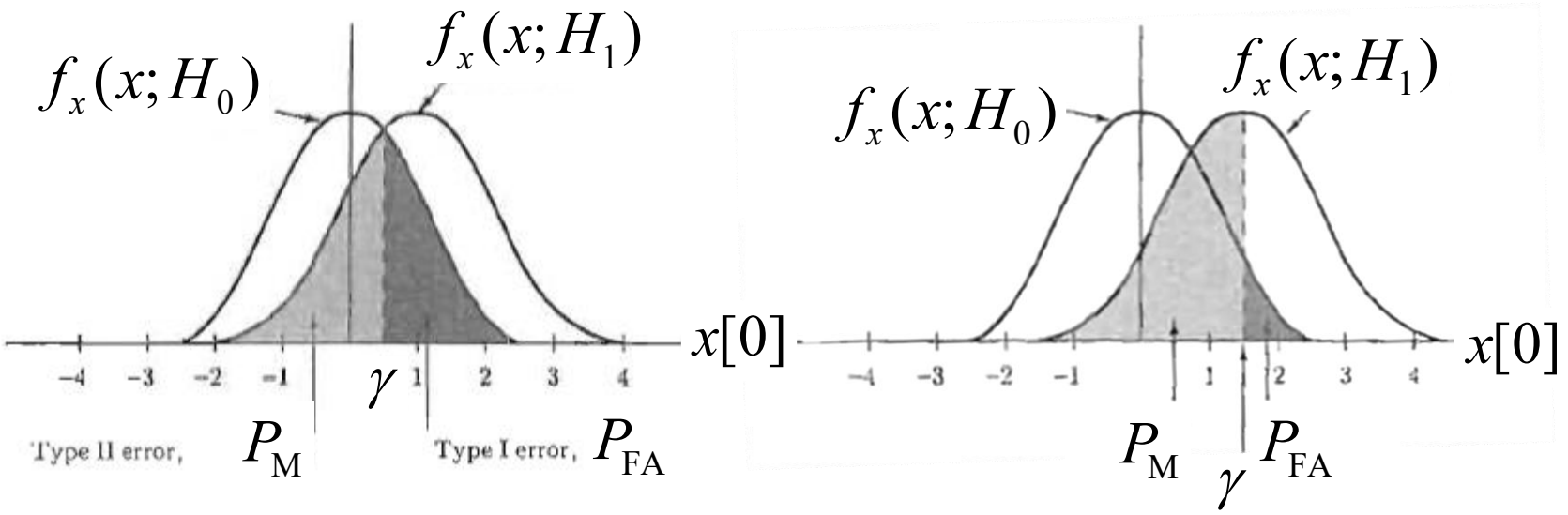
$$P_{\mathrm{M}} = \Pr\{\hat{H}_0; H_1\} \qquad P_{\mathrm{FA}} = \Pr\{\hat{H}_1; H_0\}$$

- The probabilities are: $P_{\mathrm{M}} = \Pr\{\hat{H}_0; H_1\} = \int_{R_0} f_x(x;H_1)dx$

$$P_{\mathrm{FA}} = \Pr\{\hat{H}_1; H_0\} = \int_{R_1} f_x(x;H_0)dx$$

Binary hypothesis testing

- By adjusting the threshold, **it is not possible to reduce both error probabilities**



- By increasing the threshold ➔ the probability of false alarm reduces, but the probability of miss detection increases

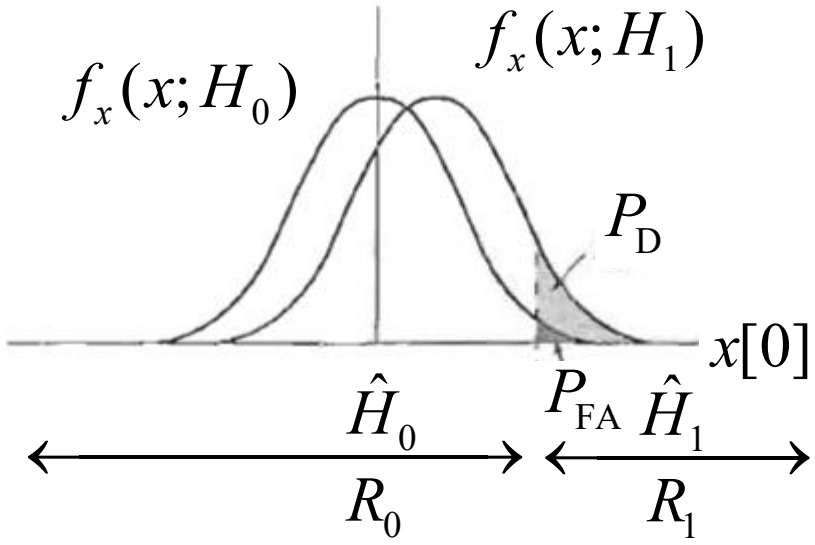


1.2

Neyman-Pearson Criterion

Neyman-Pearson criterion

- It is reasonable to minimize one type of error and keep the probability of the other error as constant



$$P_D = \int_{R_1} f_x(x;H_1)dx$$

$$P_{FA} = \int_{R_1} f_x(x;H_0)dx$$

- Neyman-Pearson criterion ➔ maximizes the probability of detection $P_D = 1 - P_M$ for a fixed probability of false alarm $P_{FA} = \alpha$



Neyman-Pearson theorem

- The detector that maximizes the probability of detection for a fixed probability of false alarm α is the **likelihood ratio test** (LRT) compared with a threshold γ chosen to satisfy the false alarm constraint

- LRT ➔ $$T(x) = L(x) = \frac{f_x(x; H_1)}{f_x(x; H_0)} > \gamma$$

- Threshold satisfies $$P_{\text{FA}} = \int_{R_1(\gamma)} f_x(x; H_0) dx = \alpha$$

Proof of Neyman-Pearson theorem

- Neyman-Pearson criterion ➔ $\max P_{\mathrm{D}} \quad \text{s.t.} \quad P_{\mathrm{FA}} = \alpha$

- Method of Lagrange multipliers

$$F(\lambda) = P_{\mathrm{D}} + \lambda(P_{\mathrm{FA}} - \alpha)$$

$$= \int_{R_1} f_x(x; H_1)dx + \lambda\left[\int_{R_1} f_x(x; H_0)dx - \alpha\right]$$

$$= \int_{R_1} \left[f_x(x; H_1) + \lambda f_x(x; H_0)\right]dx - \lambda\alpha$$

$$\Rightarrow f_x(x; H_1) + \lambda f_x(x; H_0) > 0$$

$$\Rightarrow \frac{f_x(x; H_1)}{f_x(x; H_0)} > -\lambda = \gamma$$

Example: LRT for DC level with AWGN

- Hypotheses ➔ $\boxed{\begin{aligned} H_0 &: \; x[n] = w[n] \\ H_1 &: \; x[n] = A + w[n] \end{aligned}}$ ➔ $\boxed{\begin{aligned} H_0 &: \; \boldsymbol{\mu} = \mathbf{0}_N \\ H_1 &: \; \boldsymbol{\mu} = A\mathbf{1}_N \end{aligned}}$

- LRT is $L(\mathbf{x}) = \dfrac{f_x(\mathbf{x};\boldsymbol{\mu} = A\mathbf{1})}{f_x(\mathbf{x};\boldsymbol{\mu} = \mathbf{0})} = \dfrac{\dfrac{1}{(2\pi\sigma_w^2)^{N/2}} \exp\left[-\dfrac{1}{2\sigma_w^2}\sum_{n=0}^{N-1}(x[n]-A)^2\right]}{\dfrac{1}{(2\pi\sigma_w^2)^{N/2}} \exp\left[-\dfrac{1}{2\sigma_w^2}\sum_{n=0}^{N-1}x^2[n]\right]}$

$$= \frac{\exp\left[-\dfrac{1}{2\sigma_w^2}\left(\sum_{n=0}^{N-1}x^2[n] - 2A\sum_{n=0}^{N-1}x[n] + NA^2\right)\right]}{\exp\left[-\dfrac{1}{2\sigma_w^2}\sum_{n=0}^{N-1}x^2[n]\right]}$$

$$= \exp\left[\frac{A}{\sigma_w^2}\sum_{n=0}^{N-1}x[n] - \frac{NA^2}{2\sigma_w^2}\right]$$

Example: LRT for DC level with AWGN

- But ➔ $$\exp\left[\frac{A}{\sigma_w^2}\sum_{n=0}^{N-1}x[n]-\frac{NA^2}{2\sigma_w^2}\right]>\gamma$$

$$\frac{A}{\sigma_w^2}\sum_{n=0}^{N-1}x[n]-\frac{NA^2}{2\sigma_w^2}>\ln\gamma$$

$$\frac{1}{N}\sum_{n=0}^{N-1}x[n]>\frac{\sigma_w^2}{NA}\ln\gamma+\frac{A}{2}=\gamma'$$

- Hence the Neyman-Pearson detector compares the sample mean with a modified threshold γ'

- $\frac{1}{N}\sum_{n=0}^{N-1}x[n]$ is Gaussian with mean 0 (under H_0) or A (under H_1), with variance σ_w^2/N ➔

Example: LRT for DC level with AWGN

- Therefore ➔ $P_{\mathrm{FA}} = \Pr\left\{ \frac{1}{N}\sum_{n=0}^{N-1} x[n] > \gamma' \; ; H_0 \right\} = Q\left(\frac{\gamma'}{\sqrt{\sigma_w^2 / N}} \right)$

➔ $\gamma' = \sqrt{\frac{\sigma_w^2}{N}} Q^{-1}\left(P_{\mathrm{FA}}\right)$ ➔

$$P_{\mathrm{D}} = \Pr\left\{ \frac{1}{N}\sum_{n=0}^{N-1} x[n] > \gamma' \; ; H_1 \right\} = Q\left(\frac{\gamma' - A}{\sqrt{\sigma_w^2 / N}} \right) = Q\left(Q^{-1}(P_{\mathrm{FA}}) - \sqrt{\frac{NA^2}{\sigma_w^2}} \right)$$

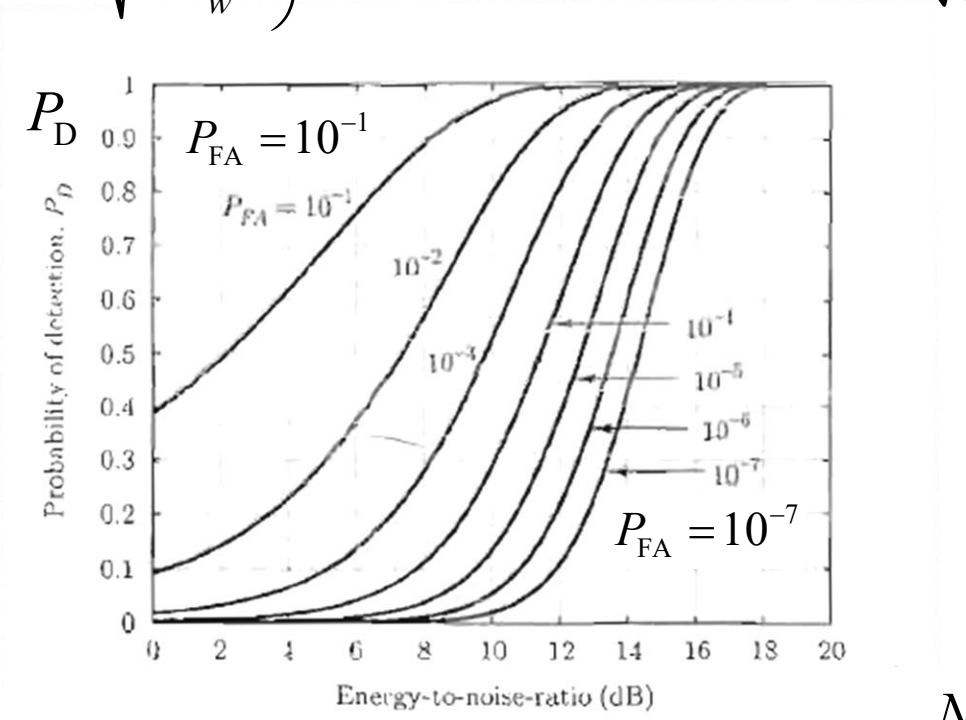
- If probability of false alarm increases ➔ threshold reduces ➔ probability of detection increases

- $d^2 = \frac{(A-0)^2}{\sigma_w^2 / N} = \frac{NA^2}{\sigma_w^2}$ deflection coefficient or SNR

Example: LRT for DC level with AWGN

$$P_{\mathrm{D}} = Q\left(Q^{-1}(P_{\mathrm{FA}}) - \sqrt{\frac{NA^2}{\sigma_w^2}} \right)$$

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{+\infty} e^{-u^2/2} du$$



$$10\log_{10}\frac{NA^2}{\sigma_w^2}$$

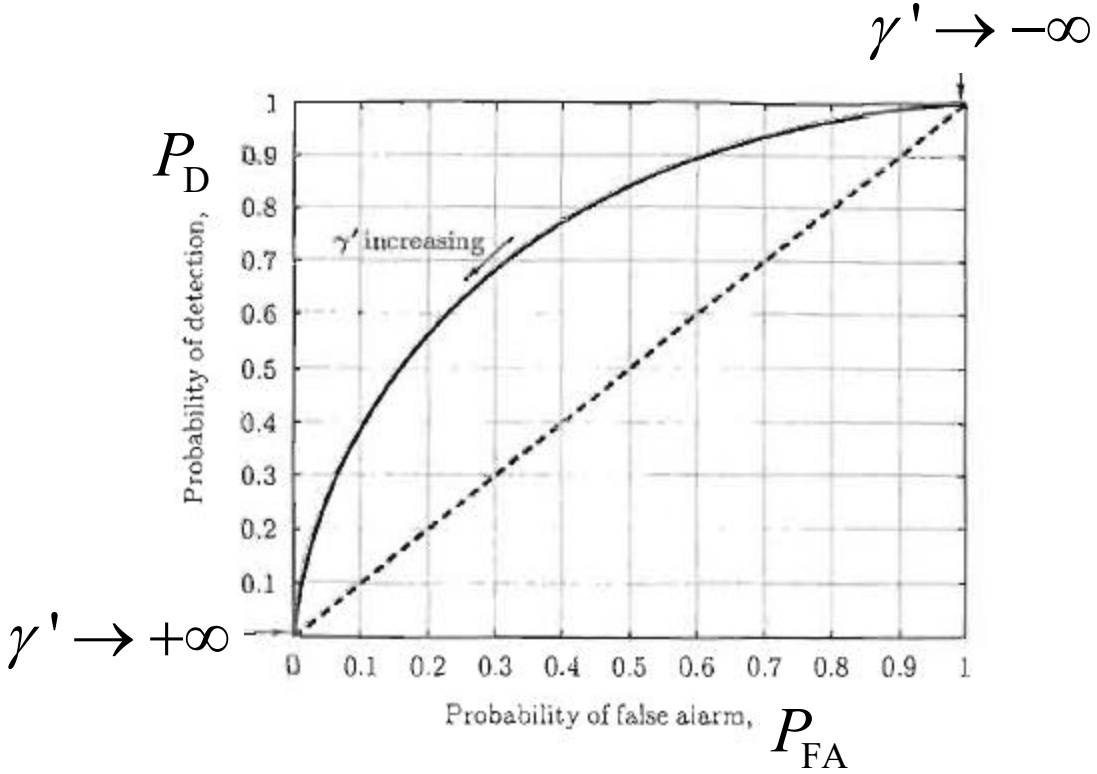


1.3

Receiver Operating Characteristic (ROC)

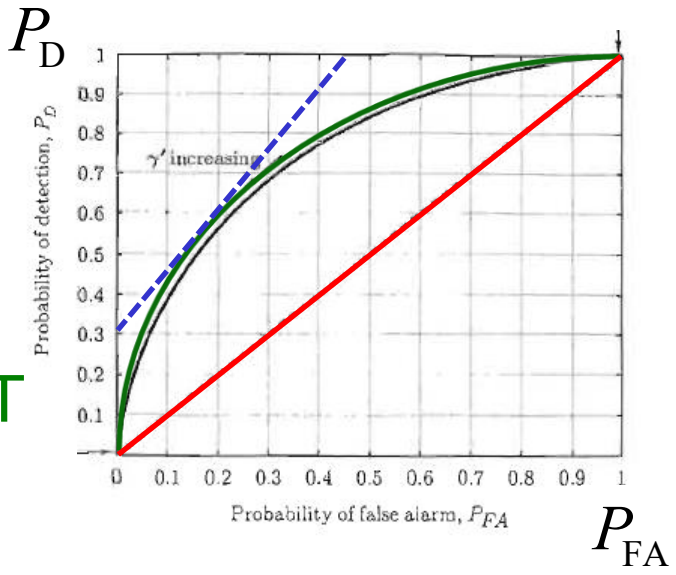
Receiver operating characteristic (ROC)

- The ROC curve plots the probability of detection as a function of the probability of false alarm, for all the possible values of the threshold



Properties of the ROC

- The ROC is nondecreasing
- The worst ROC is $P_{\mathrm{D}} = P_{\mathrm{FA}}$
- The best ROC is that of the LRT
- The ROC of the LRT is concave



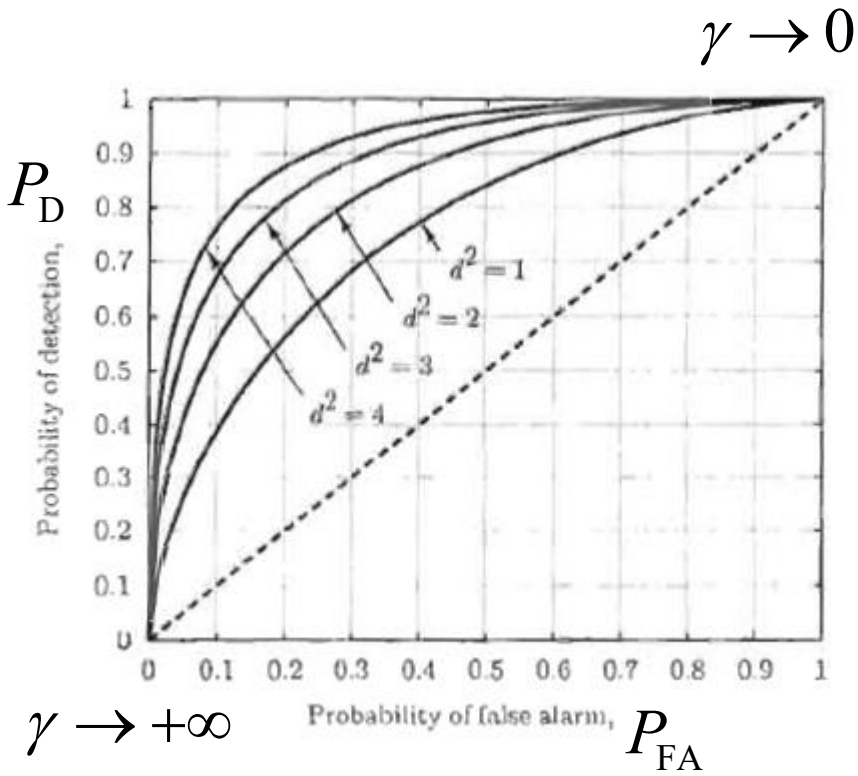
- The slope of the ROC of the LRT is the threshold γ
  ➔ the tangent in $(P_{\mathrm{FA}}, P_{\mathrm{D}}) = (\alpha, \beta)$ is $P_{\mathrm{D}} = \gamma P_{\mathrm{FA}} + \beta - \gamma\alpha$

$$\frac{\partial P_{\mathrm{D}}}{\partial P_{\mathrm{FA}}} = \gamma$$

Example: ROC of the LRT (DC level with AWGN)

- $P_{\mathrm{D}} = Q\left(Q^{-1}(P_{\mathrm{FA}}) - \sqrt{d^2}\right)$

- $d^2 = \dfrac{NA^2}{\sigma_w^2}$



- The ROC increases with the deflection coefficient
  ➔ performance improves with the sample size

Proof of $\frac{\partial P_{\mathrm{D}}}{\partial P_{\mathrm{FA}}} = \gamma$ for the LRT

- LRT $l = L(x) = \dfrac{f_x(x;H_1)}{f_x(x;H_0)}$
- $E\{l^n;H_1\} = \displaystyle\int \frac{f_x{}^n(x;H_1)}{f_x{}^n(x;H_0)} f_x(x;H_1)dx = \int \frac{f_x{}^n(x;H_1)}{f_x{}^n(x;H_0)} \frac{f_x(x;H_0)}{f_x(x;H_0)} f_x(x;H_1)dx$

  $= \displaystyle\int \frac{f_x{}^{n+1}(x;H_1)}{f_x{}^{n+1}(x;H_0)} f_x(x;H_0)dx = E\{l^{n+1};H_0\}$
- $\displaystyle\int l^n f_l(l;H_1)dl = \int l^{n+1} f_l(l;H_0)dl \Rightarrow f_l(l;H_1) = l\, f_l(l;H_0) \Rightarrow \frac{f_l(l;H_1)}{f_l(l;H_0)} = l$
- $\left.\begin{aligned} P_{\mathrm{D}} &= \displaystyle\int_{\gamma}^{+\infty} f_l(l;H_1)dl \Rightarrow \frac{\partial P_{\mathrm{D}}}{\partial \gamma} = -f_l(\gamma;H_1) \\ P_{\mathrm{FA}} &= \displaystyle\int_{\gamma}^{+\infty} f_l(l;H_0)dl \Rightarrow \frac{\partial P_{\mathrm{FA}}}{\partial \gamma} = -f_l(\gamma;H_0) \end{aligned}\right\} \frac{\partial P_{\mathrm{D}}}{\partial P_{\mathrm{FA}}} = \frac{\partial P_{\mathrm{D}}}{\partial \gamma}\frac{\partial \gamma}{\partial P_{\mathrm{FA}}} = \frac{-f_l(\gamma;H_1)}{-f_l(\gamma;H_0)} = \gamma$



1.4

Minimum Probability of Error



Probability of error

- Bayesian approach ➔ known $\Pr\{H_0\}$ and $\Pr\{H_1\}$
- Probability of error

$$
\begin{aligned}
P_{\mathrm{e}} &= \Pr\{\hat{H}_0, H_1\} + \Pr\{\hat{H}_1, H_0\} \\
&= \Pr\{\hat{H}_0 \mid H_1\}\Pr\{H_1\} + \Pr\{\hat{H}_1 \mid H_0\}\Pr\{H_0\} \\
&= P_{\mathrm{M}}\Pr\{H_1\} + P_{\mathrm{FA}}\Pr\{H_0\}
\end{aligned}
$$

- Weighted average of the probabilities of miss detection and false alarm ➔ the weights are the probabilities of the corresponding hypotheses



Minimum probability of error

- $$P_{\mathrm{e}} = (1-P_{\mathrm{D}})\Pr\{H_1\} + P_{\mathrm{FA}}\Pr\{H_0\}$$

$$= \left(1 - \int_{R_1} f_x(x \mid H_1)dx\right)\Pr\{H_1\} + \left(\int_{R_1} f_x(x \mid H_0)dx\right)\Pr\{H_0\}$$

$$= \Pr\{H_1\} - \int_{R_1}\left[f_x(x \mid H_1)\Pr\{H_1\} - f_x(x \mid H_0)\Pr\{H_0\}\right]dx$$

- Hence $f_x(x \mid H_1)\Pr\{H_1\} > f_x(x \mid H_0)\Pr\{H_0\}$ or equivalently the critical region R_1 is given by

$$L(x) = \frac{f_x(x \mid H_1)}{f_x(x \mid H_0)} > \frac{\Pr\{H_0\}}{\Pr\{H_1\}} = \gamma$$

➔ the detector with minimum probability of error is the LRT with threshold $\gamma = \Pr\{H_0\}/\Pr\{H_1\}$



Minimum probability of error

- Since $\Pr\{H_i \mid x\} = \dfrac{f_x(x \mid H_i)\Pr\{H_i\}}{f_x(x)}$, the condition

  $f_x(x \mid H_1)\Pr\{H_1\} > f_x(x \mid H_0)\Pr\{H_0\}$ is equivalent to

  $$\Pr\{H_1 \mid x\} > \Pr\{H_0 \mid x\}$$

  ➔ maximum a posteriori probability (MAP) detector

- When $\Pr\{H_0\} = \Pr\{H_1\}$ ➔ $f_x(x \mid H_1) > f_x(x \mid H_0)$ ➔ MAP detector becomes the maximum likelihood (ML) detector ➔ this is the LRT with threshold equal to 1

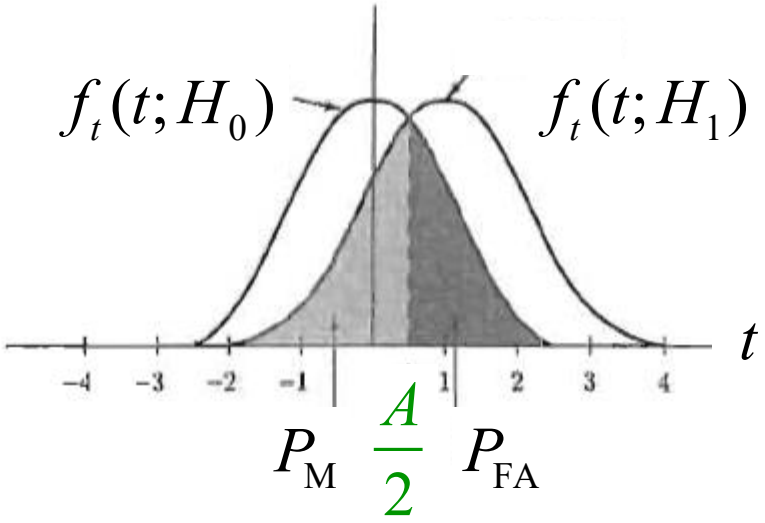
  $$L(x) = \frac{f_x(x \mid H_1)}{f_x(x \mid H_0)} > 1$$

Example: minimum probability of error (DC level with AWGN)

- $\begin{aligned} H_0 &: \; x[n] = w[n] \\ H_1 &: \; x[n] = A + w[n] \end{aligned}$ ➔ $\begin{aligned} H_0 &: \; \boldsymbol{\mu} = \mathbf{0}_N \\ H_1 &: \; \boldsymbol{\mu} = A\mathbf{1}_N \end{aligned}$ $\Pr\{H_0\} = \Pr\{H_1\} = \frac{1}{2}$

- LRT > 1 ➔ $L(\mathbf{x}) = \frac{f_x(\mathbf{x};\boldsymbol{\mu} = A\mathbf{1})}{f_x(\mathbf{x};\boldsymbol{\mu} = \mathbf{0})} = \exp\left[\frac{A}{\sigma_w^2}\sum_{n=0}^{N-1} x[n] - \frac{NA^2}{2\sigma_w^2}\right] > \gamma = 1$

- This corresponds to $t = T'(\mathbf{x}) = \frac{1}{N}\sum_{n=0}^{N-1} x[n] > \frac{A}{2} = \gamma'$



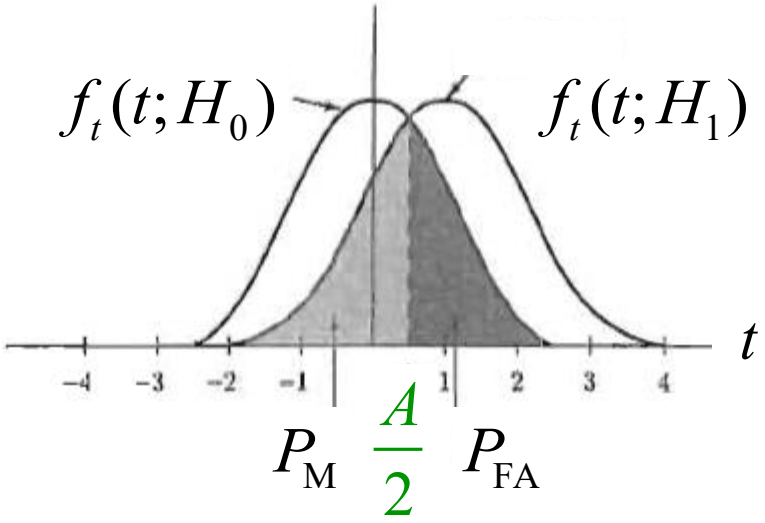
➔ The threshold γ' is chosen such that $P_M = P_{FA}$

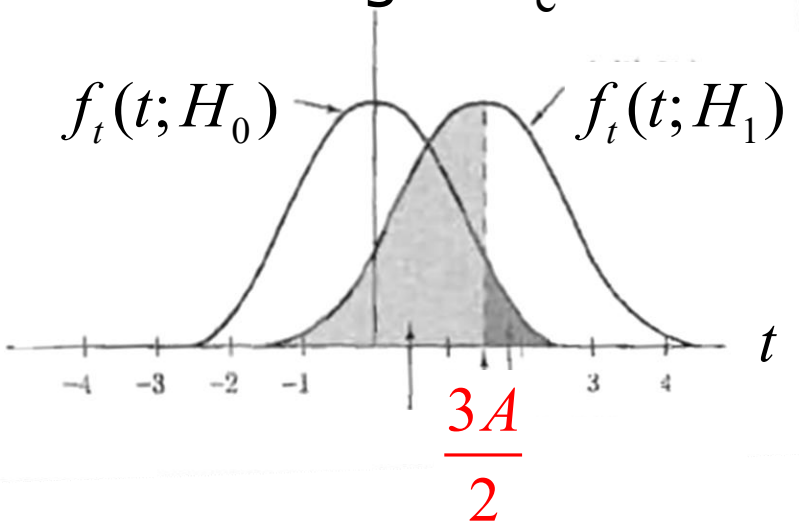
Example: minimum probability of error (DC level with AWGN)

- $P_{\mathrm{M}} = P_{\mathrm{FA}}$ ➔ $P_{\mathrm{e}} = P_{\mathrm{M}} \Pr\{H_1\} + P_{\mathrm{FA}} \Pr\{H_0\} = \dfrac{P_{\mathrm{M}} + P_{\mathrm{FA}}}{2} = P_{\mathrm{FA}}$
- Hence the minimum probability of error is

$$P_{\mathrm{e}} = P_{\mathrm{FA}} = Q\left(\frac{A}{2}\sqrt{\frac{N}{\sigma_w^2}}\right)$$

- Other thresholds would produce a larger P_{e}







1.5

Bayes Risk



Bayes risk

- The Bayes risk is a Bayesian detection approach that assign given costs to each type of error
- We define C_{ij} the cost of deciding $\hat{H} = H_i$ when the true hypothesis is H_j
- The Bayes risk is the average cost, given by

$$R = \sum_{i=0}^{1}\sum_{j=0}^{1} C_{ij} \Pr\{\hat{H}_i, H_j\} = \sum_{i=0}^{1}\sum_{j=0}^{1} C_{ij} \Pr\{\hat{H}_i \mid H_j\} \Pr\{H_j\}$$

- Usually there is no cost for a correct decision ➔ $C_{00} = C_{11} = 0$



Minimum Bayes risk

- The goal is to minimize the Bayes risk

$$
\begin{aligned}
R = {} & C_{00}\Pr\{\hat{H}_0 \mid H_0\}\Pr\{H_0\} + C_{10}\Pr\{\hat{H}_1 \mid H_0\}\Pr\{H_0\} + \\
& + C_{01}\Pr\{\hat{H}_0 \mid H_1\}\Pr\{H_1\} + C_{11}\Pr\{\hat{H}_1 \mid H_1\}\Pr\{H_1\} = \\
& = C_{00}(1 - P_{\mathrm{FA}})\Pr\{H_0\} + C_{10}P_{\mathrm{FA}}\Pr\{H_0\} + \\
& + C_{01}(1 - P_D)\Pr\{H_1\} + C_{11}P_{\mathrm{D}}\Pr\{H_1\} = \\
& = K + (C_{10} - C_{00})P_{\mathrm{FA}}\Pr\{H_0\} - (C_{01} - C_{11})P_D\Pr\{H_1\}
\end{aligned}
$$

where $K = C_{00}\Pr\{H_0\} + C_{01}\Pr\{H_1\}$ is constant

- Assuming that $C_{10} > C_{00}$ and $C_{01} > C_{11}$ ➔ we have

Minimum Bayes risk

$$R = K + (C_{10} - C_{00})\left(\int_{R_1} f_x(x \mid H_0)dx\right)\Pr\{H_0\} - (C_{01} - C_{11})\left(\int_{R_1} f_x(x \mid H_1)dx\right)\Pr\{H_1\} =$$

$$= K + \int_{R_1}\left[f_x(x \mid H_0)(C_{10} - C_{00})\Pr\{H_0\} - f_x(x \mid H_1)(C_{01} - C_{11})\Pr\{H_1\}\right]dx$$

- Hence $f_x(x \mid H_0)(C_{10} - C_{00})\Pr\{H_0\} - f_x(x \mid H_1)(C_{01} - C_{11})\Pr\{H_1\} < 0$ or equivalently the critical region R_1 is given by

$$L(x) = \frac{f_x(x \mid H_1)}{f_x(x \mid H_0)} > \frac{(C_{10} - C_{00})\Pr\{H_0\}}{(C_{01} - C_{11})\Pr\{H_1\}} = \gamma$$

➔ the detector with minimum Bayes risk is the LRT with threshold γ as expressed above



Minimum Bayes risk

- The probability of error is a special case of Bayes risk with $C_{00} = C_{11} = 0$ and $C_{10} = C_{01} = 1$ ➔ in this case, the detector that minimizes the Bayes risk is identical to the detector that minimizes the probability of error

- The minimum Bayes risk detector is equivalent to

$$\frac{\Pr\{H_1 \mid x\}}{\Pr\{H_0 \mid x\}} = \frac{f_x(x \mid H_1)\Pr\{H_1\}}{f_x(x \mid H_0)\Pr\{H_0\}} > \frac{C_{10} - C_{00}}{C_{01} - C_{11}}$$

➔ this is a generalization of the MAP detector



1.6

Multiple Hypothesis Testing



Multiple hypothesis testing

- A Bayesian detector for testing $M > 2$ hypotheses may use a Bayes risk defined as

$$R = \sum_{i=0}^{M-1} \sum_{j=0}^{M-1} C_{ij} \Pr\{\hat{H}_i \mid H_j\} \Pr\{H_j\}$$

with costs $C_{ij} > C_{jj}$, $\forall i \neq j$

- A special but interesting case is when $C_{jj} = 0$ and $C_{ij} = 1$, $\forall i \neq j$ ➔ the Bayes risk coincides with the probability of error ➔ the minimization leads to the MAP detector that decides $\hat{H} = H_k$ if

$$\Pr\{H_k \mid x\} > \Pr\{H_i \mid x\} \quad , \quad \forall i \neq k$$



Example: multiple DC levels with AWGN

- $$\begin{aligned} H_0 &: \; x[n] = -A + w[n] \\ H_1 &: \; x[n] = w[n] \\ H_2 &: \; x[n] = A + w[n] \end{aligned} \;\Rightarrow\; \begin{aligned} H_0 &: \; \boldsymbol{\mu} = A_0 \mathbf{1}_N = -A\mathbf{1}_N \\ H_1 &: \; \boldsymbol{\mu} = A_1 \mathbf{1}_N = 0\mathbf{1}_N \\ H_2 &: \; \boldsymbol{\mu} = A_2 \mathbf{1}_N = A\mathbf{1}_N \end{aligned} \qquad \Pr\{H_i\} = \frac{1}{3}$$

- MAP detector ➔ $\max_{i\in\{0,1,2\}} \Pr\{H_i \mid \mathbf{x}\} = \max_{i\in\{0,1,2\}} \dfrac{f_{\mathbf{x}|H_i}(\mathbf{x} \mid H_i)\Pr\{H_i\}}{f_{\mathbf{x}}(\mathbf{x})}$

- H_i equally likely ➔ MAP corresponds to ML

$$\max_{i\in\{0,1,2\}} f_{\mathbf{x}|H_i}(\mathbf{x} \mid H_i) = \max_{i\in\{0,1,2\}} f_{\mathbf{x}|\boldsymbol{\mu}}(\mathbf{x} \mid \boldsymbol{\mu} = A_i \mathbf{1}_N)$$

- The detector must partition the space into three regions: R_0, R_1, R_2 (one for each hypothesis)



Example: multiple DC levels with AWGN

$$\max f_{\mathbf{x}|\boldsymbol{\mu}}(\mathbf{x}\,|\,\boldsymbol{\mu}=A_i\mathbf{1}_N)=\frac{1}{(2\pi\sigma_w^2)^{N/2}}\exp\left[-\frac{1}{2\sigma_w^2}\sum_{n=0}^{N-1}(x[n]-A_i)^2\right]$$

$$\min\sum_{n=0}^{N-1}(x[n]-A_i)^2=\sum_{n=0}^{N-1}x^2[n]-2A_i\sum_{n=0}^{N-1}x[n]+NA_i^2$$

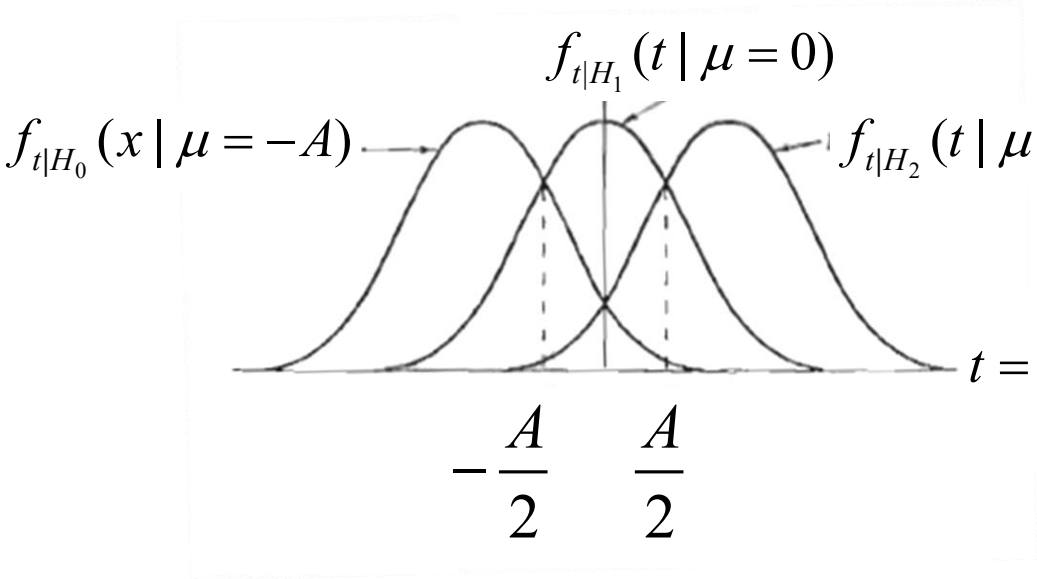
$$\min\left[-2A_i\left(\frac{1}{N}\sum_{n=0}^{N-1}x[n]\right)+A_i^2\right]$$

$$\min\left[\left(\frac{1}{N}\sum_{n=0}^{N-1}x[n]\right)^2-2A_i\left(\frac{1}{N}\sum_{n=0}^{N-1}x[n]\right)+A_i^2\right]=\left[\left(\frac{1}{N}\sum_{n=0}^{N-1}x[n]\right)-A_i\right]^2$$

- The detector decides for the hypothesis $\hat{H}=H_i$ such that the corresponding A_i is the closest one to the sample mean

Example: multiple DC levels with AWGN

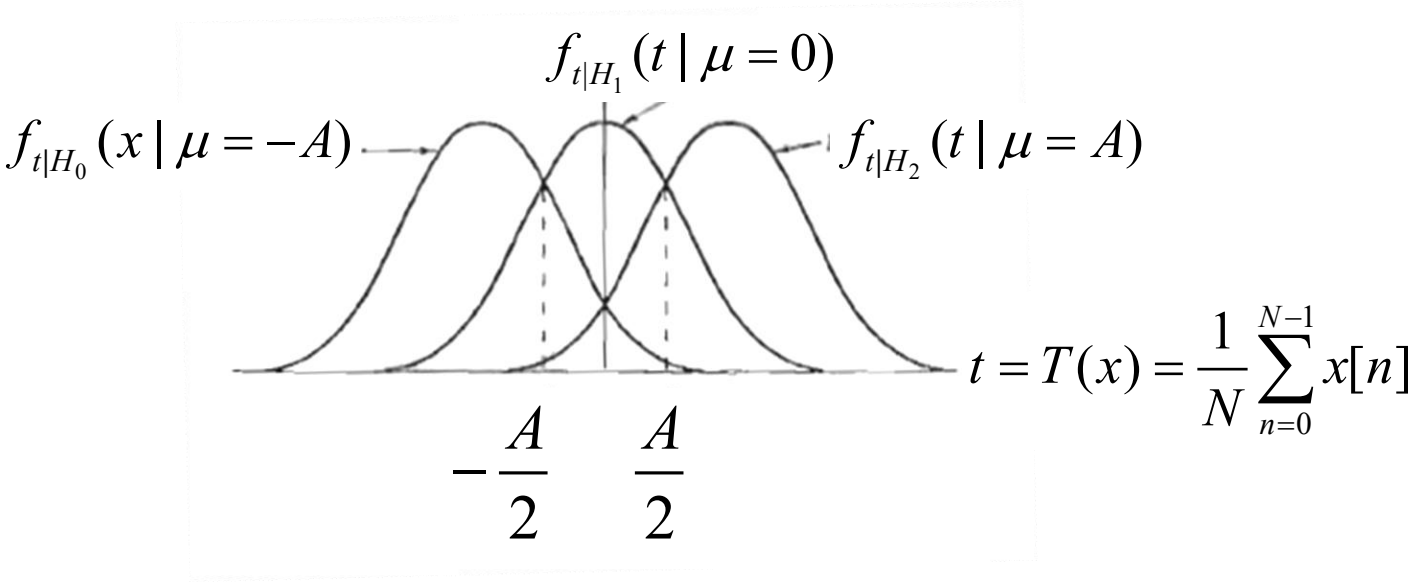
- There are two thresholds ➔ $\pm A/2$
- The three regions are $\begin{cases} R_0 = \{x: \quad T(x) < -A/2\} \\ R_1 = \{x: \quad -A/2 < T(x) < A/2\} \\ R_2 = \{x: \quad T(x) > A/2\} \end{cases}$



Example: multiple DC levels with AWGN

- The error probability is

$$P_{\mathrm{e}} = \sum_{j=0}^{2} \Pr\{\hat{H} \neq H_j \mid H_j\} \Pr\{H_j\} = \frac{1}{3} \sum_{j=0}^{2} \Pr\{\hat{H} \neq H_j \mid H_j\}$$

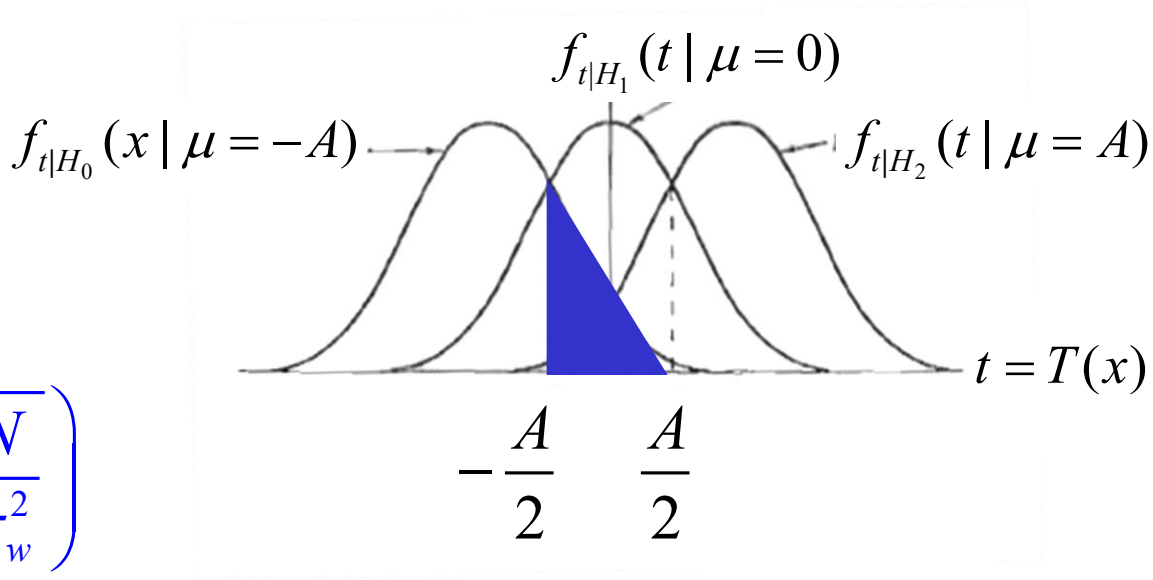


Example: multiple DC levels with AWGN

- The error probability is

$$P_e = \sum_{j=0}^{2} \Pr\{\hat{H} \neq H_j \mid H_j\} \Pr\{H_j\} = \frac{1}{3} \sum_{j=0}^{2} \Pr\{\hat{H} \neq H_j \mid H_j\}$$

$$= \frac{1}{3}\left[\Pr\{\hat{H} \neq H_0 \mid H_0\} + \Pr\{\hat{H} \neq H_1 \mid H_1\} + \Pr\{\hat{H} \neq H_2 \mid H_2\} \right]$$



$$t = T(x) = \frac{1}{N} \sum_{n=0}^{N-1} x[n]$$

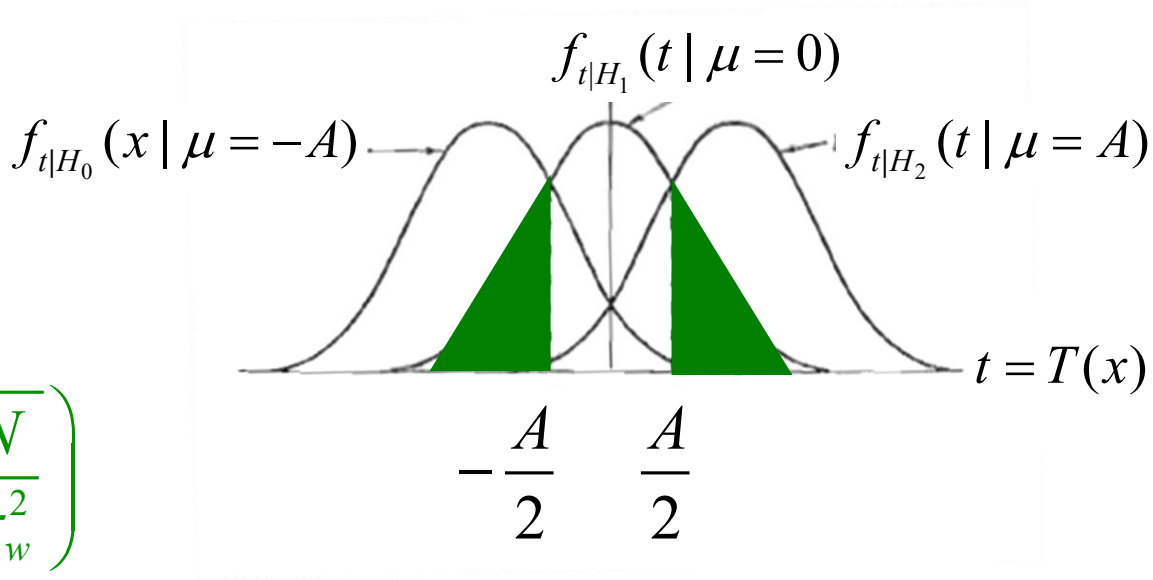
$$Q\left(\frac{A}{2} \sqrt{\frac{N}{\sigma_w^2}} \right)$$

Example: multiple DC levels with AWGN

- The error probability is

$$P_e = \sum_{j=0}^{2} \Pr\{\hat{H} \neq H_j \mid H_j\} \Pr\{H_j\} = \frac{1}{3} \sum_{j=0}^{2} \Pr\{\hat{H} \neq H_j \mid H_j\}$$

$$= \frac{1}{3}\left[\Pr\{\hat{H} \neq H_0 \mid H_0\} + \Pr\{\hat{H} \neq H_1 \mid H_1\} + \Pr\{\hat{H} \neq H_2 \mid H_2\} \right]$$



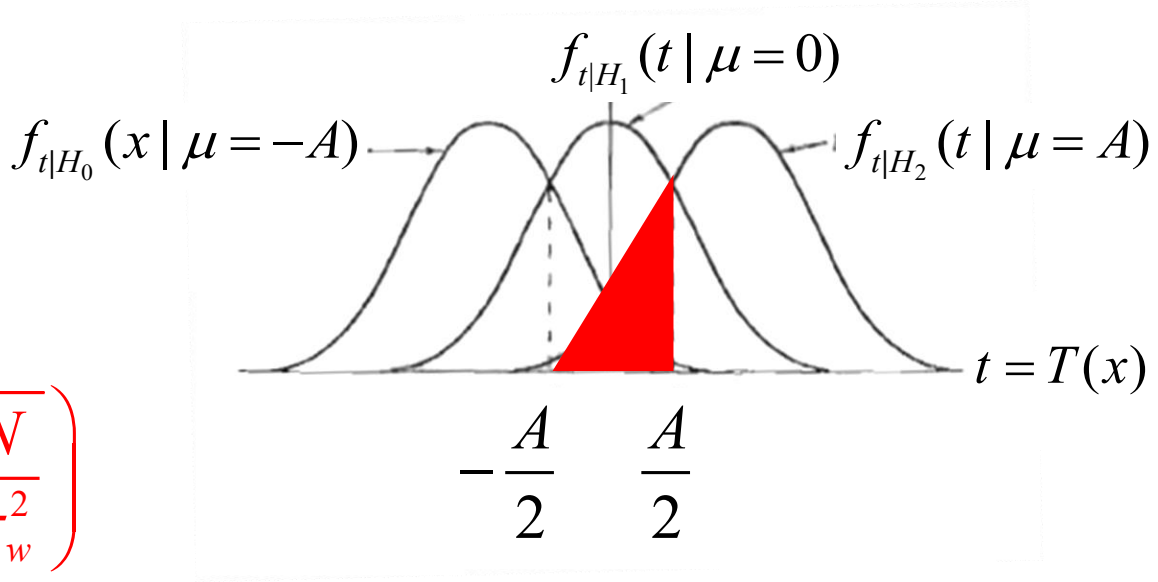
$$2Q\left(\frac{A}{2}\sqrt{\frac{N}{\sigma_w^2}}\right)$$

Example: multiple DC levels with AWGN

- The error probability is

$$P_e = \sum_{j=0}^{2} \Pr\{\hat{H} \neq H_j \mid H_j\} \Pr\{H_j\} = \frac{1}{3} \sum_{j=0}^{2} \Pr\{\hat{H} \neq H_j \mid H_j\}$$

$$= \frac{1}{3}\left[\Pr\{\hat{H} \neq H_0 \mid H_0\} + \Pr\{\hat{H} \neq H_1 \mid H_1\} + \Pr\{\hat{H} \neq H_2 \mid H_2\} \right]$$



$$t = T(x) = \frac{1}{N} \sum_{n=0}^{N-1} x[n]$$

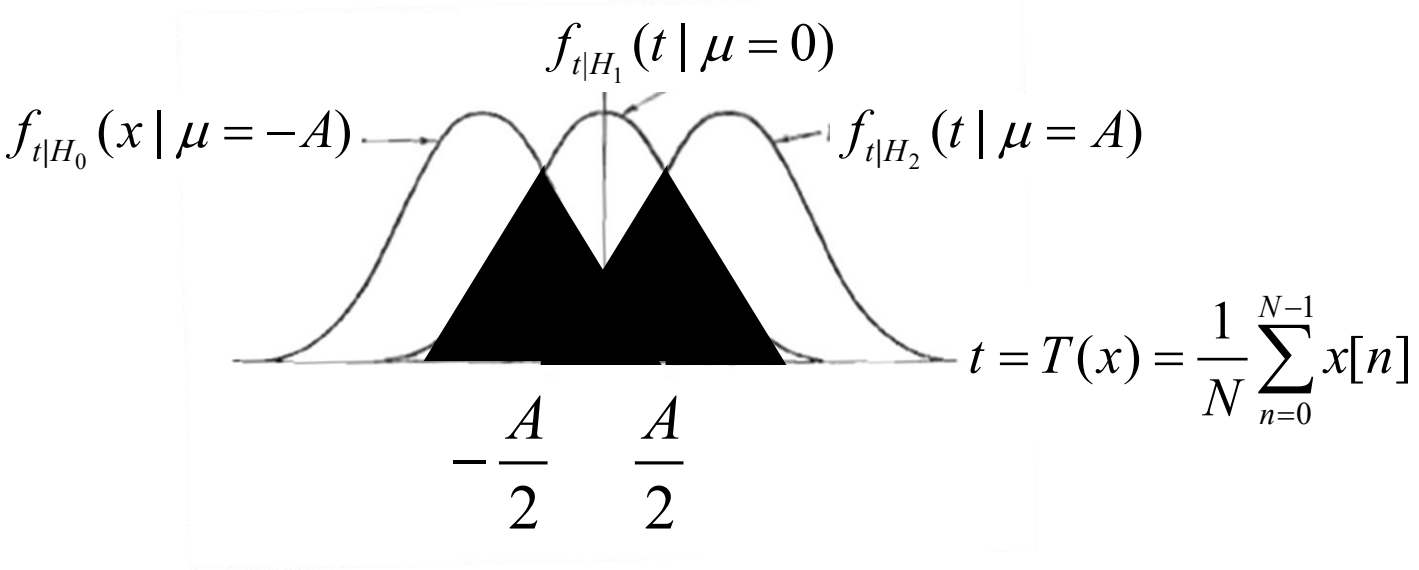
$$Q\left(\frac{A}{2} \sqrt{\frac{N}{\sigma_w^2}} \right)$$

Example: multiple DC levels with AWGN

- The error probability is

$$P_e = \sum_{j=0}^{2} \Pr\{\hat{H} \neq H_j \mid H_j\} \Pr\{H_j\} = \frac{1}{3} \sum_{j=0}^{2} \Pr\{\hat{H} \neq H_j \mid H_j\}$$

$$= \frac{1}{3}\left[Q\left(\frac{A}{2}\sqrt{\frac{N}{\sigma_w^2}}\right) + 2Q\left(\frac{A}{2}\sqrt{\frac{N}{\sigma_w^2}}\right) + Q\left(\frac{A}{2}\sqrt{\frac{N}{\sigma_w^2}}\right)\right] = \boxed{\frac{4}{3} Q\left(\frac{A}{2}\sqrt{\frac{N}{\sigma_w^2}}\right)}$$





1.7

Generalized Likelihood Ratio Test (GLRT)

Composite hypothesis testing

- Composite hypothesis testing: the pdfs of the data are partially unknown ➔ in the parametric case, we have $f_x(x;\theta_0,H_0)$ and $f_x(x;\theta_1,H_1)$ where either θ_0 or θ_1 (or both) are unknown
- If the LRT does not depend on the unknown parameter ➔ use the LRT
- If the LRT depends on the unknown parameters ➔ generalized likelihood ratio test (GLRT)
- If statistic knowledge of the unknown parameters is available ➔ LRT with Bayesian approach

Generalized likelihood ratio test (GLRT)

- The GLRT approach has two steps

- First step ➔ evaluate of the LRT, which is function of the unknown parameters θ_0 and θ_1

$$L(x) = \frac{f_x(x;\theta_1,H_1)}{f_x(x;\theta_0,H_0)}$$

- Second step ➔ replace the unknown parameters with the MLE values $\hat{\theta}_0 = \hat{\theta}_{0,\mathrm{MLE};H_0}$ and $\hat{\theta}_1 = \hat{\theta}_{1,\mathrm{MLE};H_1}$ ➔ the GLRT is:

$$L_{\mathrm{G}}(x) = \frac{f_x(x;\hat{\theta}_1,H_1)}{f_x(x;\hat{\theta}_0,H_0)}$$



Generalized likelihood ratio test (GLRT)

- The GLRT has no optimality properties, but it works quite well in practice
- The GLRT provides information about the unknown parameters, because the first step involves the estimation of the unknown parameters
- The GLRT is quite easy to apply, because the MLE is one of the easiest estimation methods
- The GLRT can be simplified when one of the hypotheses has no unknown parameters

Example: GLRT for unknown DC level with AWGN

- Hypotheses ➔ $\begin{aligned} H_0 &: \; x[n] = w[n] \\ H_1 &: \; x[n] = A + w[n] \end{aligned}$ ➔ $\begin{aligned} H_0 &: \; \boldsymbol{\mu} = \mathbf{0}_N \\ H_1 &: \; \boldsymbol{\mu} = A\mathbf{1}_N \end{aligned}$ $A \neq 0$

- MLE is $\hat{A}_{\text{MLE}} = \dfrac{\mathbf{1}_N^T \mathbf{x}}{N} = \dfrac{1}{N}\sum_{n=0}^{N-1} x[n]$

- GLRT is $L_{\text{G}}(\mathbf{x}) = \dfrac{f_x(\mathbf{x}; \boldsymbol{\mu} = \hat{A}_{\text{MLE}}\mathbf{1})}{f_x(\mathbf{x}; \boldsymbol{\mu} = \mathbf{0})} = \dfrac{\exp\left[-\dfrac{1}{2\sigma_w^2}\sum_{n=0}^{N-1}(x[n] - \hat{A}_{\text{MLE}})^2\right]}{\exp\left[-\dfrac{1}{2\sigma_w^2}\sum_{n=0}^{N-1} x^2[n]\right]}$

$$= \exp\left[\frac{\hat{A}_{\text{MLE}}}{\sigma_w^2}\sum_{n=0}^{N-1} x[n] - \frac{N\hat{A}_{\text{MLE}}^2}{2\sigma_w^2}\right] = \exp\left[\frac{N\hat{A}_{\text{MLE}}^2}{2\sigma_w^2}\right] > \gamma \quad \Rightarrow \quad \hat{A}_{\text{MLE}}^2 > \gamma'$$

LRT with Bayesian approach

- If the pdfs of the unknown parameters $f_{\theta_0}(\theta_0; H_0)$ and $f_{\theta_1}(\theta_1; H_1)$ are known ➔ Bayesian approach applied to LRT for composite hypothesis testing
- LRT ➔ $$\frac{f_x(x; H_1)}{f_x(x; H_0)} = \frac{\int f_x(x \mid \theta_1; H_1) f_{\theta_1}(\theta_1; H_1) d\theta_1}{\int f_x(x \mid \theta_0; H_0) f_{\theta_0}(\theta_0; H_0) d\theta_0} > \gamma$$
- Integration is required ➔ complexity can be large
- If $\Pr\{H_0\}$ and $\Pr\{H_1\}$ are available ➔ MAP ➔

$$\frac{f_x(x \mid H_1)\Pr\{H_1\}}{f_x(x \mid H_0)\Pr\{H_0\}} > 1 \quad \Rightarrow \quad \frac{\int f_x(x \mid \theta_1, H_1) f_{\theta_1}(\theta_1 \mid H_1) d\theta_1}{\int f_x(x \mid \theta_0, H_0) f_{\theta_0}(\theta_0 \mid H_0) d\theta_0} > \frac{\Pr\{H_0\}}{\Pr\{H_1\}}$$



Final Summary

Summary of detectors

- Binary hypothesis tests with parametric model
- If the **prior pdf** of the two hypotheses **is not available** ➔ choose a classical approach
  - Simple hypotheses ➔ Neyman-Pearson criterion
  - Composite hypotheses ➔ GLRT
- If **prior pdf** of the two hypotheses **is available** ➔ a Bayesian approach is preferable
  - Simple hypotheses ➔ minimum Bayes risk (or MAP or ML)
  - Composite hypotheses ➔ Bayesian LRT